Design of a High-Voltage Low-Ripple Converter With High-Frequency Dickson Multipliers

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***Abstract*—High-voltage, low-ripple dc-dc converters are essential in various industrial and scientific applications. Increased switching frequency helps improve converter power density but challenges circuit design. In this article, we propose a design method based on a detailed model of Dickson voltage multipliers to account for non-idealities in high-frequency multiplier circuitry. We present the design of a high-frequency 80 W, 48-V-to-7-kV dc-dc power converter with less than 10 Vpp voltage ripple. The circuit utilizes GaN-based high-frequency inverters, multiport air-core planar printed circuit board (PCB) transformers, Dickson voltage multipliers, and ripple cancellation. The PCB transformers are constructed with shielding layers to mitigate noise-induced voltage ripple. The implemented prototype can maintain a 0.1% ripple ratio while achieving 86% peak efficiency.**

***Index Terms*—gallium nitride (GaN) high-electron-mobility transistors (HEMTs), high voltage, planar transformer, ripple cancellation, voltage multiplier.**

I. Introduction

DC-DC converters with an output ranging from a few kilovolts to beyond a hundred kilovolts play an essential role in various medical and industrial applications. Many of these converters require a voltage gain of more than 100 as the available input voltage is usually much lower than the required high voltage output [2], [3]. High-voltage step-up dc-dc converters often consist of an inverter, a transformer, and a rectifier or multiplier [4]. Hence, they tend to have lower power densities, and many recent studies focus on reducing the size and weight of high-voltage converters [5], [6]. Moreover, specialized applications such as X-ray excitation [7], ultrahigh vacuum generation [8], and electric propulsion [4] require an output voltage ripple of no more than a few tenths of a percent, as a more significant ripple can adversely affect the system's performance.

An effective way to suppress the output voltage ripple is to use large filtering passive components in the rectifier stage. This inevitably increases the volume and weight of converters, lowering the power density and adding cost [6]. Previous studies have proposed hybrid circuit structures to optimize the sizes of passive components needed in multipliers and increase power density [9]. However, this approach requires a trade-off between component size and voltage stress, undermining the effectiveness of power density improvement. Alternatively, a high switching frequency allows much easier ripple filtering in which smaller passive components are needed. Previous studies have demonstrated low-ripple converters with high power density by pushing the frequency up to sub-megahertz [6], [8]. However, when the frequency goes beyond 1 MHz, parasitic capacitances in the voltage multiplier can significantly impact converter operation, resulting in deteriorated voltage regulation and increased reactive power [4], [10]. Since conventional multiplier modeling does not account for such parasitic capacitances, researchers have implemented different RC models to help with the high-frequency converter design [4], [11]. These models aim to derive an equivalent RC input impedance of multipliers that can be used for designing the transformer resonant tank. While they capture multipliers' input characteristics, the worsening voltage regulation and output ripple caused by parasitic capacitances remain ignored. In this article, we demonstrate a Dickson multiplier model that gives good estimates of the electrical behavior of the circuit and accurately predicts the output voltage and ripple.

Another approach to reducing ripple involves combining the time-shifted output of multiple converters so as to cancel the output ripple [6], [12]. Previous studies have demonstrated successful ripple cancellation at either low frequency or light load [12], [13], [14]. However, many designs tend to add extra circuitry and lower the power density at a much higher cost [12], [14]. Moreover, the implementation of this approach is challenging at high frequencies since circuit and layout parasitics can affect the outputs of the converters, preventing them from achieving effective ripple cancellation. Besides, high-frequency common-mode electromagnetic interference (EMI) caused by large current or voltage swings undermines the benefits of interleaving and causes

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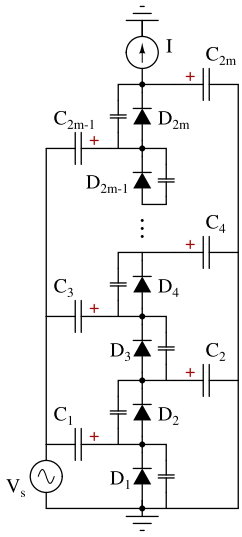


Fig. 1. Schematic of an m-stage Dickson voltage multiplier with parasitic capacitance of diodes.

excessive ripples at the output. In this article, we demonstrate a three-winding printed circuit board (PCB) transformer with embedded EMI shielding for ripple cancellation. The PCB transformer helps control parasitics and maintain system symmetry with minimum added circuitry while keeping the cost low.

We present a novel detailed quantitative study on how diodes' parasitic capacitances impact the input impedance, the voltage gain, and the output ripple of voltage multipliers. The proposed methodology helps guide the design of high-frequency, high-voltage converters to achieve high power density. Combined with ripple cancellation and EMI shielding, this article's implemented high-voltage dc-dc converter operates at 1.79 MHz, delivering 80 W at 7 kV dc voltage with a peak efficiency of about 86% and an output ripple of 0.1%. Table V compares our prototype with existing related research, indicating promising power density and voltage ripple improvements

For the organization of this article, Section II describes the high-frequency parasitic model of Dickson multipliers. Section III explains the design and operation of the proposed converter. Section IV shows the experiment data and Section V concludes the article.

II. High-Frequency Parasitics Model of Dickson Voltage Multiplier

Compared with the Cockcroft–Walton or the Hybrid multipliers, Dickson multipliers with the same bypass capacitance have better voltage regulation and less ripple [9]. Hence, in applications requiring very small ripple, the Dickson multiplier is the best alternative for improving power density. Fig. 1 shows the schematic of a m-stage Dickson multiplier, where V_s represents the peak input AC voltage with frequency f, while I is the output dc current. Our model assumes the conduction characteristic of the diodes as ideal, having a constant forward voltage drop V_D, and accounts for the diode capacitance by adding a constant capacitance C_j in parallel as shown in Fig. 1. Although the junction capacitance of diodes is nonlinear with voltage, we can obtained an equivalent constant C_j by averaging the junction charge versus voltage following the method outlined in Appendix A. We assume the ESRs of diodes and capacitors are negligible, and the leakage currents of diodes are small. The circuit's output charge per period is defined as

$$q = \frac{I}{f}. \tag{1}$$

Conventional analysis of this circuit renders the average output voltage and peak-to-peak voltage ripple as [9], [15]

$$V_{\text{out}} = 2mV_s - q\left(\sum_{k=1}^{2m}\frac{1}{C_k} - \frac{1}{2C_{2m}}\right) \tag{2a}$$

$$\Delta v_{\text{out}} = \frac{q}{C_{2m}} \tag{2b}$$

which predicts that the voltage ripple is inversely proportional to the frequency, and V_{out} approaches $2mV_s$ with increasing frequency. A detailed derivation using the approach from [12] is shown in Appendix B. However, this is not the case when considerable parasitic capacitances are present, since a voltage divider is formed between the diodes' parasitic capacitance and the bypass capacitors.

To derive a parasitic model of the Dickson multiplier that can better predict the output voltage and its ripple, we first recognize that the circuit operates in a cycle of four phases described as follows ($k = 1, ..., m$). We will focus on the average output voltage and its ripple without the need to determine the duration of each phase.

Phase 1: D_{2k-1} starts turning on, D_{2k} remains off.

Phase 2: All diodes are off, the circuit becomes a passive network formed by all parasitic and bypass capacitors.

Phase 3: D_{2k-1} turns off, D_{2k} starts turning on.

Phase 4: All diodes are off, the circuit becomes a passive network formed by all parasitic and bypass capacitors.

Fig. 2 shows the waveform of a two-stage Dickson multiplier as an example. As shown in the figure, the transitions can be complicated and all diodes do not necessarily turn on or off (when the voltage step happens) at the same time. However, it is worth noting that the capacitor voltages reach their maximum or minimum values at about the same time. Hence, to solve for the dc output voltage and its ripple, we rely on averaged circuit modeling instead of transient calculations.

We denote the voltage across capacitor C_i as V_i with the polarity defined in Fig. 1; the maximum reverse voltage blocked by diode D_i as $V_{J,i}$ (from cathode to anode) ($i = 1, ..., 2m$).

At the end of phase 1 (t_1 in Fig. 2), when the input voltage $V_{\text{in}} = -V_s$, capacitors C_{2k-1} are charged to their maximum voltages while C_{2k} (except C_{2m}) are discharged to their minimum voltages. Applying KVL to the circuit renders

$$V_{2k-1,\max} = \begin{cases} V_s - V_D & k = 1 \\ V_{2k-2,\min} + V_s - V_D & k \geq 2 \end{cases}. \tag{3}$$

For the output capacitor C_{2m}, its voltage V_{2m} may not drop to the minimum at this point due to the load current. But at high frequency, we can assume that the charge transferred between C_{2m} and the parasitic capacitance of D_{2m} is considerably larger

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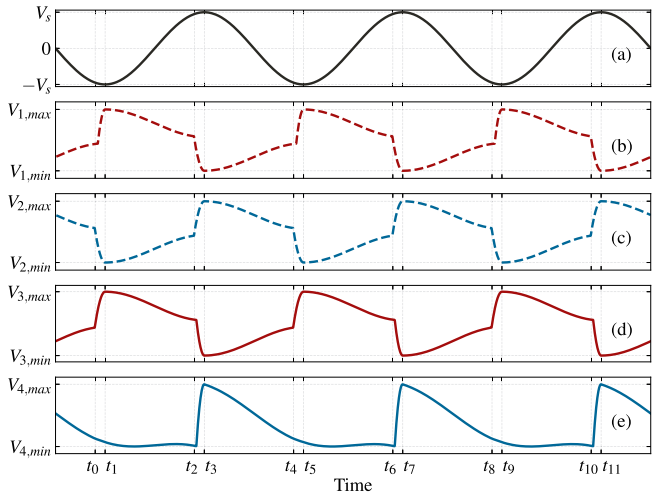


Fig. 2. Simulated idealized high-frequency waveform of a two-stage Dickson multiplier, illustrating: (a) input voltage v_s; (b) voltage across C_1; (c) voltage across C_2; (d) voltage across C_3; and (e) voltage across C_4. The waveform is plotted with $C_1 = C_2 = C_3 = 0.2C_4$ to validate the assumption that the charge transferred between C_4 and the parasitic capacitance of D_4 is considerably larger than the output charge q. The figure shows the boundary condition that during t_5 to t_6, the amount of charge transferred between C_4 and D_4's parasitic capacitance compensates the amount of charge discharged by the load current, causing V_4 to stay relatively constant during the time interval. Notice that phase transitions do not happen simultaneously at t_0, t_2,..., t_{10}.

than the output charge q, as shown in Fig. 2. Hence, V_{2m} approximately reaches the minimum when $V_{\text{in}} = -V_s$. Meanwhile, diodes D_{2k} block the maximum reverse voltage that can be expressed as

$$V_{J,2k} = \begin{cases} V_{2,\min} + V_D & k = 1 \\ V_{2k,\min} - V_{2k-2,\min} + V_D & k \geq 2 \end{cases}. \tag{4}$$

Similarly at the end of phase 3 (t_3 in Fig. 2), when $V_{\text{in}} = V_s$, capacitors C_{2k} are charged to their maximum voltages while C_{2k-1} are discharged to their minimum voltages. The relationship between these voltages can be derived as

$$V_{2k,\max} = V_{2k-1,\min} + V_s - V_D. \tag{5}$$

And the maximum reverse voltages blocked by D_{2k-1} are

$$V_{J,2k-1} = \begin{cases} V_{2,\max} + V_D & k = 1 \\ V_{2k,\max} - V_{2k-2,\max} + V_D & k \geq 2 \end{cases}. \tag{6}$$

During phase 2 to the end of phase 3 (t_1 to t_3 in Fig. 2), capacitor voltages vary between their maximum and minimum values as V_{in} changes. For any capacitor C_i ($i = 1, ..., 2m-1$), when its voltage drops from $V_{i,\max}$ to $V_{i,\min}$, its charge transfer can be concluded into three categories: 1) it charges the parasitic capacitor of D_i from $-V_D$ to $V_{J,i}$; 2) it discharges the parasitic capacitor of D_{i+1} from $V_{J,i+1}$ to $-V_D$; 3) it charges C_{i+1} by q. Fig. 3 illustrates this charge transfer with the circuit diagram. Summarizing the charge relationship renders the averaged KCL equations

$$C_i(V_{i,\max} - V_{i,\min}) = C_j(V_{J,i} + V_{J,i+1} + 2V_D) + q \tag{7}$$

where C_j is the averaged constant parasitic capacitance of diodes. During the voltage drop of C_{2m}, it only charges the

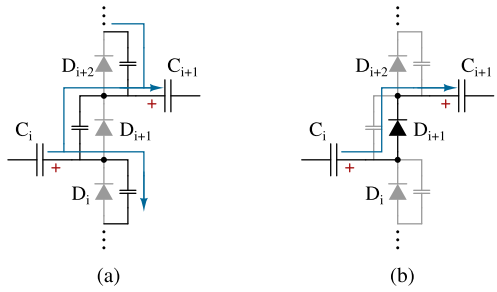


Fig. 3. Illustration of charge transfer from phase 2 to the end of phase 3. (a) After D_i and D_{i+2} turn off but before D_{i+1} turns on. C_i charges and discharges the parasitic capacitors of D_i and D_{i+1}. The charge transferred into C_{i+1} compensates its discharging cycle during phase 4 and (b) After D_{i+1} turns on, C_i charges C_{i+1} by q to compensate for the amount charge eventually goes into the load.

parasitic capacitance of D_{2m} and the load. Hence, the resulting averaged charge transfer equation is

$$C_{2m}(V_{2m,\max} - V_{2m,\min}) = C_j(V_{J,2m} + V_D) + \frac{q}{2}. \tag{8}$$

As a result, we get $4m$ KVL equations and $2m$ averaged KCL equations. Therefore, the unknown $4m$ capacitor voltages and $2m$ diode voltages can be solved using (3)–(8).

The closed-form solution to capacitor voltages can be further simplified as

$$\mathbf{v} = \mathbf{A}^{-1}\mathbf{u}(V_s, I, f) \tag{9}$$

where $\mathbf{v} \in \mathbb{R}^{4m}$ consists of the maximum and minimum voltages of all $2m$ bypassing capacitors, $\mathbf{A} \in \mathbb{R}^{4m \times 4m}$ is a constant matrix with all the capacitances and $\mathbf{u} \in \mathbb{R}^{4m}$ is determined by the input voltage amplitude V_s, output current I and frequency f. We can easily obtain the average output voltage and its peak-to-peak ripple as

$$V_{\text{out}} = (V_{2m,\max} + V_{2m,\min})/2 \tag{10a}$$

$$\Delta v_{\text{out}} = V_{2m,\max} - V_{2m,\min}. \tag{10b}$$

It is worth noting that the derivation of the model does not require a sinusoidal input. As long as the input voltage is symmetrical, the output voltage and ripple can be calculated by (9) and (10).

To derive an expression for the input impedance of the multiplier, as in [4], we assume fundamental harmonic approximation for the circuit. With a sinusoidal excitation V_s, the multiplier can be approximated as a lumped parallel RC circuit, where the resistance R_{eq} stands for the real power component while the capacitance C_{eq} consumes the reactive power as stored in both parasitic and bypassing capacitors. Given the results from (9), the input impedance can be expressed as

$$R_{eq} = \frac{V_s^2}{I(V_{2m,\max} - V_{2m,\min})} \tag{11a}$$

$$C_{eq} = \frac{1}{V_s^2}\sum_{i=1}^{2m}\left(C_i\left(\frac{\Delta V_i}{2}\right)^2 + C_j\left(\frac{V_{J,k} + V_D}{2}\right)^2\right) \tag{11b}$$

where

$$\Delta V_i = V_{i,\max} - V_{i,\min}. \tag{12}$$

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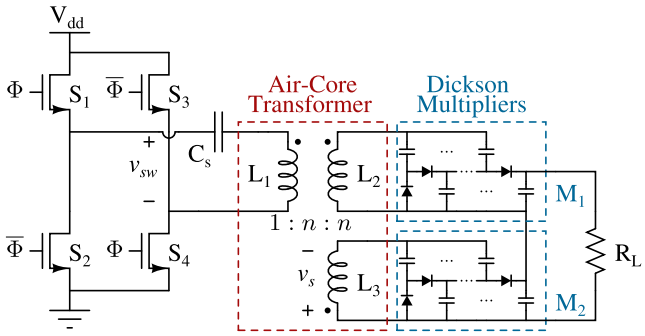


Fig. 4. Schematic of the proposed converter with a three-winding air-core transformer and two identical Dickson multipliers driven out-phased.

III. Converter Design

A. Converter Structure

Fig. 4 shows the schematic of the proposed converter, which consists of a full-bridge inverter, a three-winding air-core transformer and two Dickson multipliers. R_L represents the load powered by the high-voltage dc output, and V_{dd} represents the low-voltage dc input. We target our design to deliver 80 W continuously at 7 kV output voltage, hence R_L is approximately 600 kΩ. The Dickson multipliers M_1 and M_2 have their outputs connected in series, and are individually driven by different transformer windings. The transformer is designed such that its secondary and tertiary windings are identical, so that M_1 and M_2 can be are symmetrically driven by 180° shifted signals to cancel out their voltage ripples at the output. The self-inductances of the primary, secondary and tertiary windings are represented as L_1, L_2, and L_3, respectively. C_s is a series dc-blocking capacitor connecting the transformer and the full-bridge inverter. The transistors S_1 to S_4 are controlled by complimentary gate signals as noted in Fig. 4. Since the secondary and tertiary windings of the transformer are identical, we represent the voltage across both of them with v_s. We denote the square wave voltage generated by the full-bridge inverter as v_{sw}.

B. Design Considerations for the Voltage Multipliers

The multiplier diodes are selected with the goal of minimizing their reverse recovery loss and to have low parasitic device capacitance. Hence, the preferred choice is to use Schottky diodes as they have very little to negligible reverse recovery loss and perform well at high frequencies.

Even though one may want to avoid designs with a large number of stages, there is another trade off between transformer turns and multiplier stages under given voltage gain constraints. A small number of stages gives better voltage regulation but at the cost of more transformer turns. This can lead to greater losses and more high-frequency parasitics. Therefore, it takes a few design iterations to determine the final parameters for the converter.

As for our proposed design methodology, we first choose an initial value of multiplier stages. Relying on the multiplier model derived in Section II, we choose small bypass capacitors that are comparable to the parasitic capacitance of diodes while achieving the desired voltage ripple to increase the converter's power density. When the voltage multiplier design is determined, (9) gives the required input excitation to generate desired output voltage, i.e., AC voltage amplitude V_s and its frequency. Furthermore, (11) calculates an approximated input impedance of the multiplier that can be useful for designing the resonant tank.

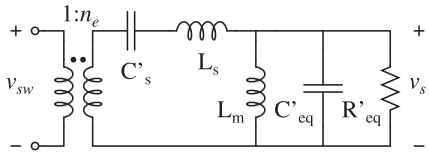


Fig. 5. Equivalent circuit of the symmetrical three-winding transformer loaded by the input impedance of voltage multipliers.

C. Transformer and Resonant Tank Design

One of the advantages of PCB transformers is their manufacturing precision and reproducibility, as it grants symmetrical secondary and tertiary windings for a three-winding transformer. In this case, the converter design can be simplified significantly by assuming identical winding parameters. In Fig. 4, since the transformer is symmetrical, the coupling between L_2 and L_1 is the same as that between L_3 and L_1. Thus, we can combine the circuit into an equivalent two-winding transformer, of which the primary side parameters remain the same as in the original circuit while the secondary (tertiary) side impedances are combined. We can then draw the equivalent circuit as in Fig. 5. To simplify the design, we assume first harmonic approximation and the voltage multiplier is represented with its input impedance as by (11). The transformer is modeled as a cantilever equivalent circuit. The cantilever model parameters are given as

$$L_m = \frac{1+k_0}{2} L_2 \tag{13a}$$

$$L_s = \left(\frac{1+k_0}{2k^2} - 1 \right) \frac{1+k_0}{2} L_2 \tag{13b}$$

$$n_e = \frac{1+k_0}{2k} \sqrt{\frac{L_2}{L_1}} \tag{13c}$$

where k is the coupling coefficient between the primary and secondary (or tertiary) windings and k_0 is the coupling coefficient between the secondary and tertiary windings. In Fig. 5, C_s' is the equivalent dc blocking capacitance from Fig. 4 referred to the secondary side of the ideal transformer, which can be given as

$$C_s' = C_s / n_e^2. \tag{14}$$

The input impedance of the multipliers is referred to the equivalent circuit as R_{eq}' and C_{eq}', and the referred impedance is half the values as calculated by (11). C_{eq}' also includes the parasitic capacitance C_w across the transformer windings as

$$R_{eq}' = R_{eq}/2, \quad C_{eq}' = 2C_{eq} + C_w. \tag{15}$$

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The voltage across R'_{eq} represents the input voltage of each voltage multiplier, of which the amplitude is V_s. As long as a voltage multiplier design is determined, R'_{eq}, C'_{eq}, V_s and the frequency of Fig. 5 can be obtained. Hence, we can design the transformer parameters to meet the voltage gain while maintaining good efficiency.

To maximize the efficiency of the converter, it is important to minimize the reactive currents in the transformer windings as it results in extra losses in the equivalent series resistance (ESR), especially in the primary winding with larger current. Hence, it is beneficial to have L_m and C'_{eq} resonate at the switching frequency, and their the resonant frequency is

$$f_r = \frac{1}{2\pi\sqrt{L_m C'_{eq}}}. \tag{16}$$

In practice, if the switching frequency is set exactly at the resonance of L_m and C'_{eq}, there might be insufficient inductive current in the circuit to discharge the output capacitances of full-bridge transistors during the switching transitions, losing zero-voltage-switching. Therefore, the resonant frequency of L_m and C_{eq} should be chosen slightly higher than the switching frequency, i.e., to present an inductive impedance at the operating frequency to enable ZVS. Detailed design processes are presented in Section IV-B.

Similarly, L_s and C'_s can be tuned to resonate at the same frequency to maximize the voltage gain. But in practice, it is desired to have enough inductive current to maintain the ZVS condition for transistors, and thus the resonance of L_s and C'_s should be chosen lower than the switching frequency. If we solely rely on L_s to achieve ZVS without adding any extra impedances, it is also possible to have a large C'_s that blocks dc current. In either case, the series resonant branch also acts as a filter to suppress third and higher harmonics in the circuit.

D. Transformer Shielding Design

As mentioned in previous sections, we use PCB transformers to better implement symmetrical secondary and tertiary windings for ripple cancellation. However, PCB transformers inherently tend to have large common mode parasitic capacitances, which are known as a major source of EMI [16]. In the case of our proposed converter shown in Fig. 4, such parasitic capacitors may provide coupling between the primary and secondary (tertiary) windings. As a result, some induced high-frequency current can be injected directly from the switching nodes into the output, increasing the voltage ripple.

Different shielding techniques have been developed to mitigate the common mode noise between windings, but few studies focus on multiport transformers with stacked outputs as in Fig. 4 [16], [17]. In this section, we propose a three-port PCB transformer design that incorporate shielding layers, which alleviate the impact of parasitic capacitors and minimizes the output voltage ripple.

We arrange the PCB layers according to Fig. 6, in which two four-layer PCBs are implemented to accommodate four shielding layers and four winding layers. Two identical primary winding layers are connected in parallel and located in the middle of the stack. The secondary and tertiary windings are then arranged at the top and bottom of the stack.



Fig. 6. Arrangement of transformer PCB layers. The transformer is implemented with two four-layer PCBs overlapping each other. Two primary layers are connected in parallel with an air gap between two PCBs.

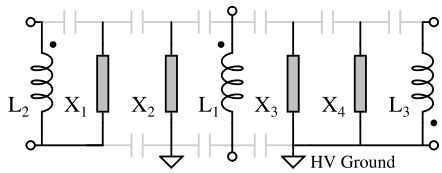


Fig. 7. Schematic showing the connections for shielding layers, which are represented as the shaded squares. The parasitic capacitance between different layers is shown in light gray.

The shielding layers X_1 to X_4 are inserted between the primary and secondary (tertiary) as shown in Fig. 6. Fig. 7 illustrates how shielding layers are connected to the transformer windings. X_2 to X_4 are connected to the high voltage ground, and X_1 is tied to the node where multipliers M_1 and M_2 are interconnected. The shielding layers are designed to have low impedance at the switching frequency. As a result, any noise current induced by switching node voltages will be guided to the high-voltage ground, thus protecting the output voltage ripple seen by the load. We use double shielding layers to further reduce their impedance.

IV. Experimental Results

A. Dickson Voltage Multiplier Model Verification

We implement three Dickson voltage multipliers with different parameters, denoted as "A", "B" and "C", to verify the modeling proposed in Section II. Multiplier A uses Si PN diodes, Multiplier B uses SiC Schottky diodes, and multiplier C uses Si Schottky diodes. All multipliers are designed to drive a 600 kΩ resistive load at 3 to 3.5 kV. Detailed circuit parameters are listed in Tables I–III. All multipliers are driven by the same high-frequency transformer that provides the associated input voltage amplitudes. The operating frequencies are chosen to avoid excessive reactive current flow in the transformer winding. Because the BAT240A diodes have a relatively low voltage rating, we connect two in series and use it as a single switching device.

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TABLE I
DESIGNED VALUES FOR VOLTAGE MULTIPLIER A

Circuit Parameter	Value and Description
m	6, number of stages
V_s	303 V, input voltage amplitude
f	1.33 MHz, input voltage frequency
D_1 to D_{12}	RFU02VSM6STR, 600 V 0.2 A Si PN diodes
C_1 to C_6	220 pF, 2 kV rated C0G capacitor
C_7 to C_8	200 pF, 3 kV rated C0G capacitor
C_9 to C_{11}	1.5 nF, 4kV rated X7R capacitor
C_{12}	1.5 nF, 6 kV rated X7R capacitor

TABLE II
DESIGNED VALUES FOR VOLTAGE MULTIPLIER B

Circuit Parameter	Value and Description
m	8, number of stages
V_s	262 V, input voltage amplitude
f	1.35 MHz, input voltage frequency
D_1 to D_{16}	GB01SLT06-214, 650 V 1 A SiC Schottky diode
C_1 to C_6	220 pF, 2 kV rated C0G capacitor
C_7 to C_{10}	200 pF, 3 kV rated C0G capacitor
C_{11} to C_{15}	1.5 nF, 4kV rated X7R capacitor
C_{16}	6 nF, 6 kV rated X7R capacitor

TABLE III
DESIGNED VALUES FOR VOLTAGE MULTIPLIER C

Circuit Parameter	Value and Description
m	9, number of stages
V_s	190 V, input voltage amplitude
f	1.45 MHz, input voltage frequency
D_1 to D_{18}	BAT240A, 240 V 0.4 A Si Schottky diodes
C_1 to C_{12}	320 pF, 3 kV rated C0G capacitor
C_{13} to C_{17}	1.5 nF, 4kV rated X7R capacitor
C_{18}	1640 pF, 5 kV rated C0G capacitor

Figs. 8 and 9 summarize the measured and calculated output voltage and ripple of multipliers comparing experiment and different models. Fig. 10 displays experimentally measured input voltage and output ripple waveform of the multipliers. According to the results, our proposed model gives a maximum 0.6% error when predicting the output dc voltage and is able to accurately estimate the output ripple with less than 15% deviation.

B. Transformer Implementation

We use multiplier C with Si Schottky diodes from Section IV as in Table III. for our converter design. According to the multiplier model, the voltage gain is approximately 16 at the nominal load condition with a switching frequency around 1.5 to 2 MHz. Similarly, the associated equivalent capacitance C'_{eq} is about 125 pF. Hence, assuming the input voltage to be 45 V and L_s is relatively small at the nominal condition, the equivalent turn ratio can be approximated as

$$n_e \approx \frac{\pi V_{\text{out}}}{16 \times 4V_{dd}} = 3.8. \tag{17}$$

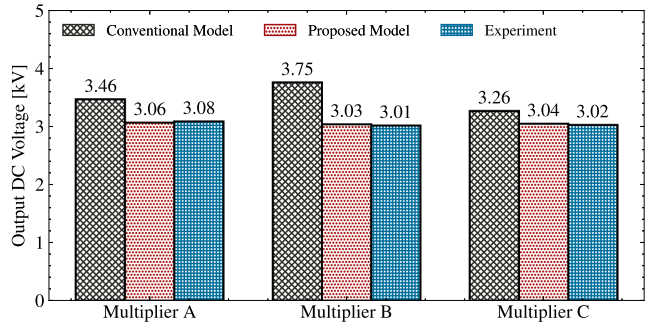


Fig. 8. Output dc voltage comparison between models and experiment for different multipliers. Conventional model calculation is based on (2a) while the proposed model uses (9) and (10a).

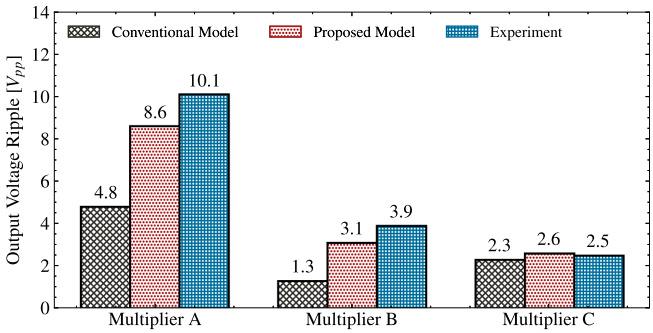


Fig. 9. Output voltage ripple comparison between models and experiment for different multipliers. Conventional model calculation is based on (2b) while the proposed model uses (9) and (10b).

To have a reasonable inductance for designing transformer geometries, the switching frequency is chosen to be $f_s \approx 1.8$ MHz, and the resonant frequency for L_m and C'_{eq} is slightly higher and determined to be $f_r \approx 2.2$ MHz. We use $C'_{eq} = 145$ pF for calculation with extra 20 pF accounting for parasitic capacitances from the PCB transformer. As a result, the needed L_m is about

$$L_m = \frac{1}{4\pi^2 f_r^2 C'_{eq}} \approx 36\ \mu\text{H}. \tag{18}$$

We then use the obtained n_e and L_m values as criteria to check the transformer layout design with the following methodology: First, we select a standard stackup provided by a PCB manufacturer, which sets the clearance between each layers. We choose the initial values of transformer diameter and turn numbers according to the analytical expressions for winding inductances given by Mohan et al. [18]. Given these geometry parameters, we then use FastHenry simulation [19] to get the inductance matrix of the transformer layout and calculate the effective turns ratio and inductances using (13). The resulting values are compared with our criteria, and geometry parameters are adjusted accordingly until we converge with analytical results.

The finalized transformer design has a 7:27 turns ratio with 8 cm outer diagonal and 3 cm inner diagonal. The inner-to-outer dimension ratio is close to 0.3, which is suggested by Lee to achieve optimal winding quality factor [20]. We implement the

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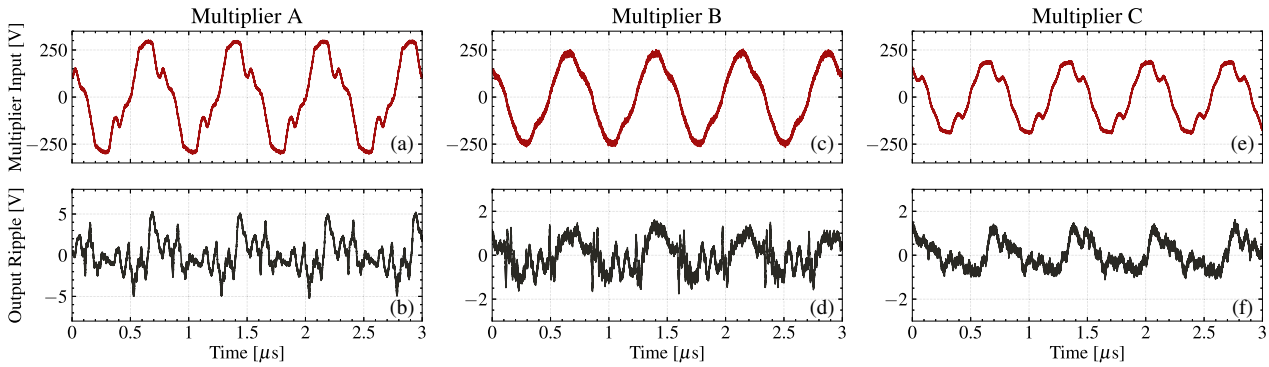


Fig. 10. Experimentally measured waveform of different multipliers. (a) & (b) Input voltage and ripple of the multiplier A with 303 V amplitude at 1.33 MHz. (c) & (d) Input voltage and ripple of the multiplier B with 262 V amplitude at 1.35 MHz and (e) & (f) Input voltage and ripple of the multiplier C with 190 V amplitude at 1.45 MHz.

TABLE IV
DESIGNED VALUES FOR PROPOSED CONVERTER

Parameter	Value and Description
S_1 to S_4	LMG5200, 80V GaN half bridge power stage
C_S	50 nF, 500 V rated C0G capacitor
L_1	2.38 μH, self-inductance of the primary layer
L_2, L_3	41.2 μH, self-inductance of secondary and tertiary
k	0.88, coupling coeff. between primary and secondary
k_0	0.765, coupling coeff. between secondary and tertiary
f_s	1.79 MHz, switching frequency

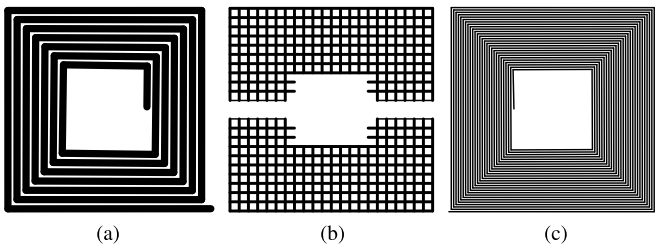


Fig. 11. Proposed layout for transformer windings and shielding layers. Separated two segments of a shielding layer are connected externally to reduce the induced power loss by the magnetic field. The dimensions of the layout are 8 cm for outer diagonal and 3 cm for inner diagonal. (a) Primary winding. (b) Shielding Layers. (c) Secondary and tertiary windings.

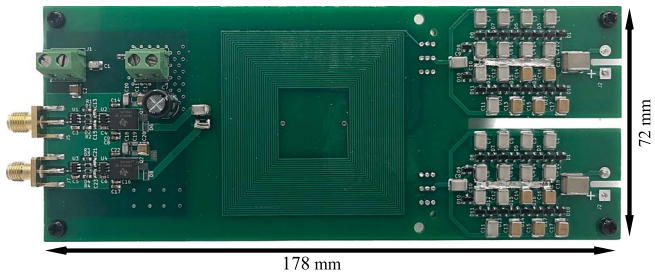


Fig. 12. Picture of the proposed converter prototype. The high-voltage part of the converter is covered by corona dope to strengthen electrical isolation.

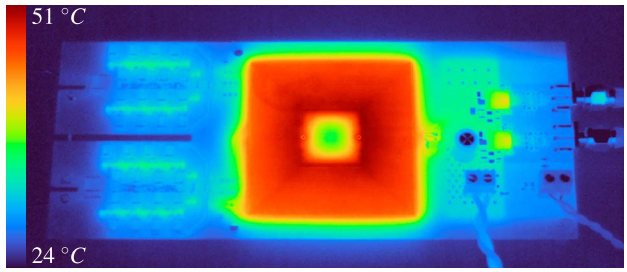


Fig. 13. Thermal picture of the proposed converter operating at 7 kV, 80 W after reaching steady state. The maximum temperature observed is 51°C when the ambient temperature is set at 24°C.

transformer with two four-layer 1.6-mm-width PCBs and set the distance between the boards to be 2 mm. The trace widths are 2 and 0.4 mm for primary and secondary (tertiary) windings, respectively. Detailed transformer parameters are listed in Table IV. Fig. 11 displays the layout for transformer windings and shielding layers. Two separate segments of a shielding layer are connected externally according to Fig. 7 to minimize the induced loss by the magnetic field.

C. Converter Implementation

Pictures of the proposed prototype are shown in Figs. 12 and 13. The converter operates at 1.79 MHz with 7 kV, 80 W output. Key components implemented in the prototype are listed in Tables I and IV. We coat the converter with a thin layer of corona dope to provide insulation and prevent unwanted discharge.

Fig. 14 summarizes the measured efficiencies of the converter across a wide power range. Including the 0.15 W gate loss for driving full-bridge transistors, the overall efficiency of the converter is about 86% when delivering 80 W at 7 kV.

Fig. 15 displays the oscilloscope-captured voltage waveform of v_{sw} across full-bridge switching nodes and v_s across the tertiary transformer winding. We use an MSO9404A scope with N2790A differential probes to measure the waveform. With the output dc voltage at 7 kV, the amplitude of v_s is about 215 V, which verifies the multiplier model prediction that the voltage gain of each multiplier is approximately 16.

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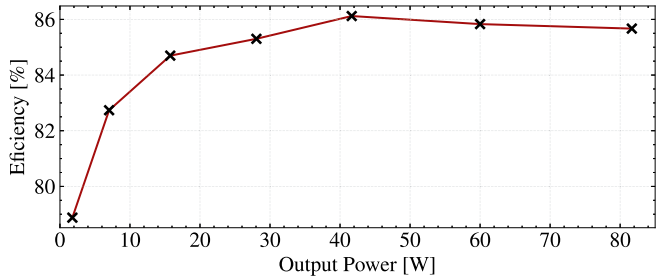


Fig. 14. Converter efficiency versus output power. The output power is regulated by adjusting the dc input voltage.

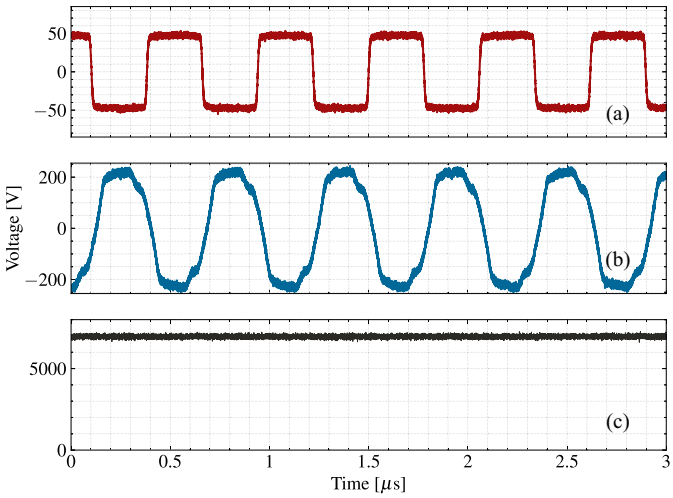


Fig. 15. Experimental voltage waveform measured at 7 kV, 80 W. (a) Switching node differential voltage v_{sw}. (b) Voltage across the tertiary winding v_s and (c) Output dc voltage measured using an 1000:1 probe. The smooth transient edges of v_{sw} prove achieved ZVS.

We use the same scope and the CalTest CT4028 high-voltage probe with AC coupling to capture the ripple waveform. As a comparison, the measured voltage ripple with shielding is shown in Fig. 16 while the ripple without shielding is in Fig. 17. The voltage ripple at lower side multiplier M_2 is slightly larger than the higher side due to the common-mode EMI from the switching nodes. Comparing both cases proves that our shielding design effectively reduces the noise and ripple. The remaining imbalance is primarily due to unshielded fringing capacitive coupling between the primary winding and the secondary winding. Nevertheless, the overall output ripple is still no larger than the ripple of M_2, achieving approximately 0.1% ripple at 7 kV, 80 W. Fig. 18 shows the transient response of implemented converter between zero input and driving full load, indicating fast rising time within 200 μs.

Table V compares the power density and ripple of our proposed converter to other recent works. Our prototype achieves the highest power density among those while achieving a very small voltage ripple ratio.

V. Conclusion

In this article, we propose the detailed design of a high-voltage dc-dc converter to achieve power density while maintaining low output ripple and high efficiency. A detailed

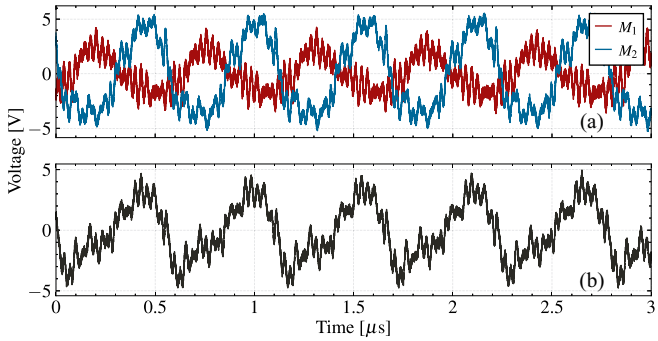


Fig. 16. Output voltage ripples measured with shielding. (a) Multiplier ripples measured at the outputs of M_1 and M_2 and (b) Overall ripple measured across the load resistor. The ripple across the multipliers is phase-shifted and generates an overall smaller ripple, verifying the effect of ripple cancellation.

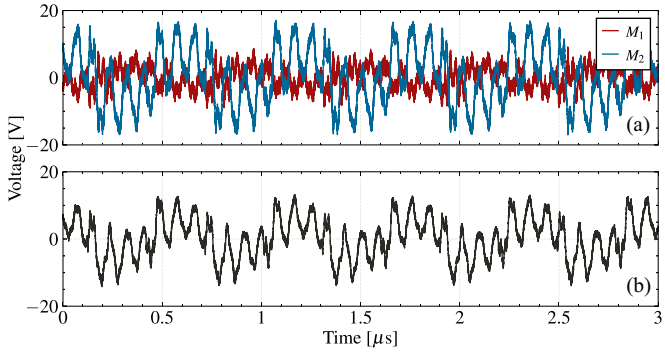


Fig. 17. Output voltage ripples measured without shielding. (a) Multiplier ripples measured at the outputs of M_1 and M_2 and (b) Overall ripple measured across the load resistor. The ripple from M_2 is significantly larger, dominating the overall ripple.

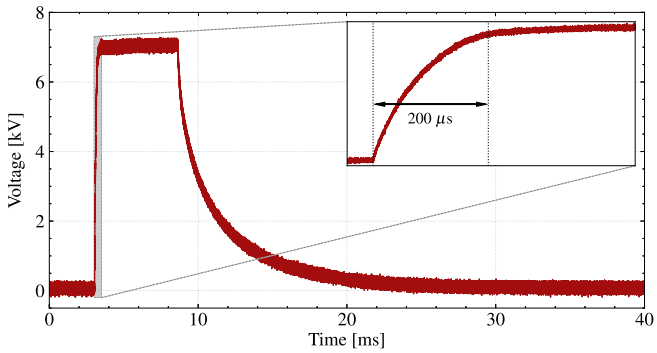


Fig. 18. Experimental pulsed output transients between zero input and driving full load. The waveform is measured when pulsing the converter at 20 Hz with 5.6 ms pulse width.

model for Dickson voltage multiplier is proposed in this article to improve high-frequency circuit design. Compared with conventional models [4], [11], [15], the proposed closed-form approach quantitatively reveals how diodes' parasitic capacitances impact the input impedance, the voltage gain, and the output ripple of voltage multipliers. The proposed prototype consists

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TABLE V
PERFORMANCE COMPARISON OF RECENTLY PUBLISHED WORK ON LOW-RIPPLE HIGH-VOLTAGE CONVERTERS

Output dc Voltage	Output Power	Switching Frequency	Peak Efficiency	Output Voltage Ripple Ratio	Power Density	Year
7 kV	**80 W**	**1.79 MHz**	**86%**	**0.13%**	**0.62 W/cm^{-3}**	**This Work**
2.5 kV	6 W	1 MHz	52%	–	0.04 W/cm^{-3}	2022 [4]
15 kV	22.5 kW	21 kHz	–	0.02%	0.03 W/cm^{-3}	2021 [14]
40 kV	565 W	500 kHz	85%	0.25%	0.56 W/cm^{-3}	2021 [6]
7 kV	350 W	500 kHz - 1.15 MHz	89%	0.85%	0.07 W/cm^{-3}	2018 [8]
20 kV	400 W	20 kHz - 40 kHz	–	0.95%	–	2017 [13]

Note: Bold entries are intended.

of a three-winding air-core transformer with shielding for ripple cancellation. The prototype is experimentally implemented and tested with 1.79 MHz switching frequency. The maximum output power is 80 W with 86% peak efficiency, and the ripple is around 0.1% at full power.

APPENDIX A
PARASITIC CAPACITANCE OF DIODES

We represent the overall parasitic capacitance of a diode as C_j. For PN diodes, it consists of three subcomponents as

$$C_j(v,i) = C_b + C_d + C_p \tag{19}$$

where C_b represents the barrier capacitance, C_d represents the diffusion capacitance, and C_p represents the parasitic capacitance from the package and layout. As related to different physical mechanisms, C_b varies with junction voltage while C_d is determined by the forward current. We assume C_p is a constant due to fixed package and layout. To estimate C_j with an equivalent constant value, we average C_b over junction voltage and approximate C_d with average load current I. Hence, the expression for equivalent constant C_j is

$$C_j = \frac{\int_0^{V_{J,\max}} C_b(v)dv - C_d(I)V_D}{V_{J,\max} + V_D} + C_p \tag{20}$$

where V_D and $V_{J,\max}$ are the forward voltage and the maximum reverse voltage across the diode. The relationship between C_b and junction voltage can be found on device datasheet, and $C_d(I)$ can be calculated as

$$C_d(I) = \frac{\tau q_0 I}{kT} \tag{21}$$

where τ is the mean lifetime of the charge carrier, k is the Boltzmann constant, q_0 is the electron charge and T is the temperature [21]. For Schottky diodes, we ignore the C_d term and the equivalent constant C_j is

$$C_j = \frac{\int_0^{V_{J,\max}} C_b(v)dv}{V_{J,\max} + V_D} + C_p. \tag{22}$$

APPENDIX B
DERIVATION OF THE CONVENTIONAL DICKSON MULTIPLIER MODEL

To derive the conventional model of an m-stage Dickson multiplier as expressed by (2), we assume ideal diodes without parasitic capacitance and the forward voltage drop. C_{2m} is discharged by q (as defined in (1)) during a period. At periodic steady state, the same amount of charge has to be transferred from C_1 to C_2, C_2 to C_3,..., C_{2m-1} to C_{2m}. For the conventional model, due to the lack of parasitic capacitors, the charge transfer happens almost instantly as the diodes turn on, and the capacitor voltages jump between maximum and minimum values accordingly as

$$C_i(V_{i,\max} - V_{i,\min}) = q \tag{23}$$

when the input voltage $V_{\text{in}} = -V_s$, D_{2k-1} turns on and D_{2k} remains off. Applying KVL to the circuit renders

$$V_{2k-1,\max} = \begin{cases} V_s & k = 1 \\ V_{2k-2,\min} + V_s & k \geq 2 \end{cases}. \tag{24}$$

Similarly, when $V_{\text{in}} = V_s$, D_{2k} turns on and D_{2k-1} remains off. Applying KVL to the circuit renders

$$V_{2k,\max} = V_{2k-1,\max} + V_s. \tag{25}$$

In summary, we have $4m$ equations for solving $2m$ maximum and $2m$ minimum voltages. The analytical solution to the average output voltage (the average voltage across C_{2m}) and output voltage ripple ($V_{2m,\max} - V_{2m,\min}$) is thus (2).

APPENDIX C
DERIVATION OF THE INPUT IMPEDANCE

To approximate the input impedance of the multiplier, we assume it consumes all the reactive power while delivering the real power to the load. With a parallel RC input impedance model, its real and reactive power can be calculated as

$$P = \frac{V_s^2}{2R_{eq}}, \quad Q = V_s^2 \omega C_{eq} \tag{26}$$

where $\omega = 2\pi f$. Equating input and output real power renders

$$P = I\frac{V_{2m,\max} + V_{2m,\min}}{2}. \tag{27}$$

Similarly, the reactive power in the multiplier capacitors sum up as the input reactive power

$$Q = \sum_{i=1}^{2m} \left(\omega C_i \left(\frac{\Delta V_i}{2} \right)^2 + \omega C_j \left(\frac{V_{J,k} + V_D}{2} \right)^2 \right). \tag{28}$$

Equation (11) can thus be derived from the equations above.

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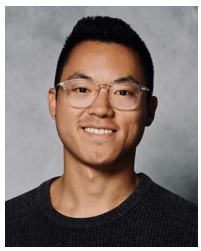
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